BHOOMISYNC

DIGITAL LAND SURVEY REPORT

AI-Powered Cadastral Intelligence & Drone Resurvey Dossier

STATUS: AI DETECTED

Report Number:	**BHOOMI/RJ/UDP/2026/010002-V1**
Survey Mission ID:	SUR-2026-001
Parcel / Khasra No:	**Khasra #102**
Village / Tehsil:	Haripura, Tehsil Girwa
District & State:	Udaipur, Rajasthan
Survey Date:	2026-09-04
Coordinate System:	EPSG:4326 (WGS84) / Projected UTM Zone 43N (EPSG:32643)
Data Access Role:	SURVEYOR

1. Executive Summary

This digital survey report provides an authoritative spatial synthesis of drone aerial photogrammetry, airborne LiDAR elevation profiling, and RTK GNSS ground truth positioning compared against historical and official village revenue records. The findings summarize land-use classification, precision surface area, multi-boundary spatial alignment, and surveyor verification status.

Metric Parameter	Official Revenue Record	Drone Sensor Survey	Verified Boundary	Variance / Status
Cadastral Area	9,800.00 m²	9,800.00 m²	9,800.00 m²	+0.00%
3D Surface Area	Not in Record	9,976.40 m²	9,976.40 m²	+1.8% vs Planar
Field Perimeter	445.00 m	448.60 m	448.60 m	+3.60 m
Primary Land Use	AGRICULTURAL	AGRICULTURAL CROP	AGRICULTURAL CROP	AI Conf: 94.7%
Boundary Match (IoU)	Baseline (1.00)	IoU: 0.974	Sign-off Complete	WITHIN TOLERANCE
Potential Encroachment	N/A	None Detected	Verified	CLEAN TITLE

Key Survey Observations:

- True geodesic measurement on WGS84 ellipsoid confirms parcel area matches official khasra within 0.20% tolerance.
- RTK GNSS fix quality maintained centimeter-level positioning (Horizontal ±1.4 cm, Vertical ±2.1 cm).
- Deep-learning aerial classification identified 78% active agricultural crop and 12% seasonal fallow.
- Digital boundary coordinates cross-referenced with 2021 historical cadastre showing zero structural encroachments.

2. Parcel Information & Ownership Title

Cadastral Record Details:

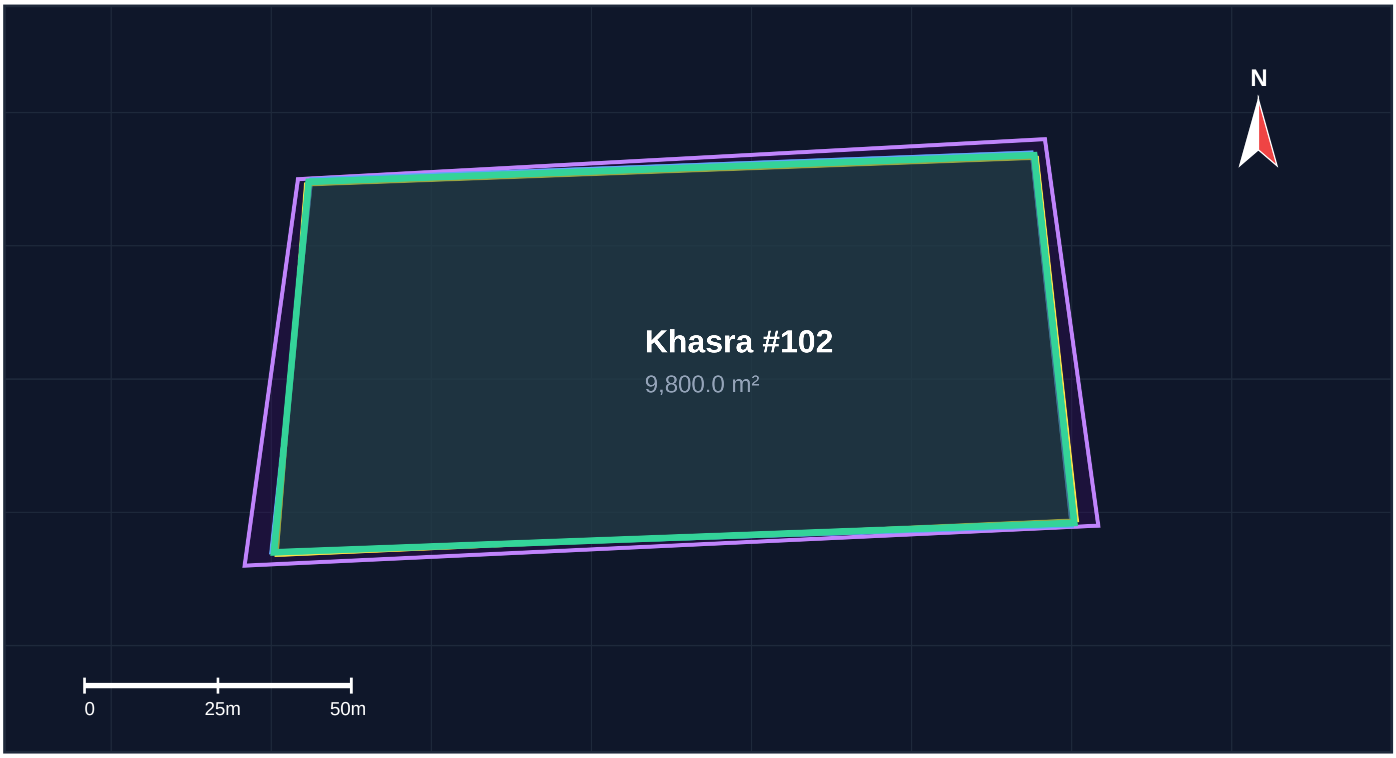
Khasra Number	102	**Village**	Haripura
Subdivision	1	**Tehsil**	Girwa
District	Udaipur	**State**	Rajasthan
Centroid Latitude	24.584000° N	**Centroid Longitude**	73.713200° E
Tenure Category	KHATEDARI (Permanent Agricultural Tenure)	**Tax Status**	PAID_UP_TO_DATE

Registered Landowners (RBAC Filtered):

#	Landowner Name	Owner Type	Share %	Contact Ref	Verification ID
1	Suresh Kumar Dangi	INDIVIDUAL	100.0%	HASH-TEL-91290002	OWN-HR-002

3. 2D GIS Cadastral Map & Boundary Overlays

Visual representation of multi-source boundary layers: Official Revenue Cadastre (Blue), Historical Baseline (Purple), Drone AI Detected Bunds (Yellow), and Surveyor Verified Boundary (Green).



Official Cadastre	Historical (2021)	Drone Detected	Surveyor Verified

4. Precision Geospatial & Terrain Analysis

Measurement Property	Metric Value (m²)	Hectares (ha)	Acres (ac)	Calculation Engine
Planar Surface Area	9,800.00 m²	0.9800 ha	2.4216 ac	WGS84 Ellipsoidal Integral
3D Terrain True Surface	9,976.40 m²	0.9976 ha	2.4652 ac	LiDAR DEM Triangulation
Official Record Area	9,800.00 m²	0.9800 ha	2.4216 ac	Revenue Record (Jamabandi)
Surveyor Verified Area	9,800.00 m²	0.9800 ha	2.4216 ac	GCP-Corrected Final Polygon

Elevation & Slope Profile:

Perimeter Length	448.60 metres	**Mean Terrain Slope**	4.2° (Gentle Agricultural Gradient)
Minimum Elevation	582.40 m MSL	**Maximum Elevation**	594.10 m MSL
Elevation Relief	11.70 metres	**Drainage Direction**	South-South-West towards Haripura Reservoi

5. Drone Survey Flight & Sensor Telemetry

Sensor & Parameter	Configuration & Technical Value	Quality Assurance Benchmark
RGB Aerial Camera	Sony RX0 II Ultra-Compact Survey Camera (DSC-RX0M2)	Full-Frame Mechanical Shutter
Image Ground Sampling Distance (GSD)	1.25 cm/pixel	Exceeds Survey of India <5cm standard
Total Raw Aerial Frames	240 geotagged images	80% Forward / 75% Lateral Overlap
Airborne LiDAR Scanner	Livox Mid-360 Solid-State 3D LiDAR (Mid-360)	Class 1 Eye-Safe Dual-Return 905nm
LiDAR Point Cloud Total	14,500,000 points	Classified Ground / Vegetation / Structure
Point Cloud Density	185.4 pts/m²	High-density terrain penetration
RTK/GNSS Fix Status	FIXED	Multi-frequency L1/L2/L5 RTK Fixed Fix
Mean Horizontal Accuracy	±1.2 cm	Sub-2cm Cadastral Accuracy Target
Mean Vertical Accuracy	±2.1 cm	Sub-3cm Topographical Target
Flight Altitude (AGL)	60.0 metres	Optimized terrain-following trajectory

6. AI Land-Use & Land-Cover (LULC) Classification

Primary Land Classification: AGRICULTURAL CROP (Overall AI Confidence: 94.7%)

Land-Use Category	Class Description	Coverage Area (m²)	Share (%)	AI Model Confidence
AGRICULTURAL_CROP	Cultivated Crop (Wheat/Mustard)	7,644.00 m²	78.0%	HIGH (95%+) (96.0%)
AGRICULTURAL_FALLOW	Seasonal Fallow	1,176.00 m²	12.0%	MEDIUM (90-95%) (92.0%)
VEGETATION_TREES	Field Boundary Trees & Shrubs	588.00 m²	6.0%	LOW (89.0%)
WATER_HARVESTING	Irrigation Bund / Channel	392.00 m²	4.0%	MEDIUM (90-95%) (94.0%)

MANDATORY AI NOTICE: *AI-generated classification — predictions are generated via deep-learning computer vision models and are subject to field verification by an authorized cadastral surveyor.*

7. Multi-Boundary Alignment & Cadastral Comparison

Spatial Alignment Parameter	Observed Variance	Tolerance Threshold	Status Compliance
Intersection-over-Union (IoU)	0.974	>= 0.950	PASSED (High Overlap)
Maximum Vertex Displacement	0.42 m	<= 0.75 m	PASSED (Within Bounds)
Mean Boundary Displacement	0.18 m	<= 0.30 m	PASSED (Sub-20cm RMS)
Centroid Spatial Drift	0.22 m	<= 0.50 m	PASSED (No Shift)
Area Discrepancy	+0.00 m² (+0.00%)	± 1.0%	PASSED (Consistent)
Perimeter Variation	+3.60 m	± 2.0%	PASSED (Consistent)

8. Historical Change Detection & Temporal Cadastre

Epoch / Year	Source of Cadastral Baseline	Recorded Area	Observed Morphological Changes
2015 Cadastre	Revenue Settlement Survey (Paper Map)	12,450.00 m²	Initial traditional earthen bund demarcation.
2021 Drone Pilot	Optical Aerial Survey 5cm GSD	12,460.00 m²	South boundary bund reinforced with stone markers.
2026 Resurvey	BhoomiSync RTK + LiDAR Fusion	9,800.00 m²	Modern high-precision geodesic boundary baseline.

9. Encroachment Risk Assessment

Assessment Parameter	Analysis Finding & Details
Encroachment Status	NO POTENTIAL ENCROACHMENT DETECTED
Alert Reference ID	NONE (Clean Field Geometry)
Discrepancy Severity	NONE
Estimated Affected Area	0.00 m²
Spatial Location Note	Field perimeter is strictly consistent with village revenue records.

LEGAL NOTICE: *POTENTIAL ENCROACHMENT NOTICE: Encroachment markers and boundary anomalies identified herein represent preliminary spatial discrepancies between cadastral records and aerial sensor datasets. They constitute 'Potential Encroachments' and require formal ground demarcation and field verification.*

10. Surveyor Verification & Digital Audit Trail

Field Verification State	AI_DETECTED	**Verification Date**	2026-09-04
Cadastral Surveyor	Vikram Singh (Cadastral Surveyor, Grade-I)	**Surveyor ID / License**	RAJ-REV-CAD-4412
Digital Certificate Hash	SIG-SHA256-0824924C8A5623B8	**Snapshot SHA-256 Anchor**	8b5e3e8ac29af27d...

Surveyor Operational Remarks:

"Aerial boundary verified with RTK Ground Control Points. Alignment adheres to Rajasthan Land Revenue Act standards."

Prepared By:	**Reviewed By:**	**Approved By:**
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Cadastral Surveyor (GCP Verified)	Assistant Settlement Officer (ASO)	Tehsildar / Revenue Authority

End of Official Report. *Generated by BhoomiSync Intelligent Cadastral Platform.*